

# MH2500: Probability and Intro to Statistics

## Tutorial 8 and Review for Quiz 2

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# Quiz 2 Information

- Topics:
  - **Continuous Random Variables - pdf, cdf and its distributions (Uniform, Normal, Exponential, Gamma), apply Normal approximation to Binomial.**
  - **Finding density function of random variable.**
  - **Joint pmf, Marginal pmf, Conditional pmf, Sum of discrete r.v. .**
  - **Joint cdf, Joint pdf, Marginal density, Sum of Continuous r.v., Conditional density, Independence of two r.v. .**
  - **Ordered Statistics.**
- Material: 2nd part of Lecture 5-2 to Lecture 10-1, Tutorials 5-8.
- Weightage: **20% of final grade.**
- Date: **Week 11, 29 October 2021, Friday.**
- Time: **9 - 10.20am.** (Report 15 mins earlier outside the venue.)
- Venue: **Exam Hall C (Group T3), LT8 (Groups T7 & T10)**
- Calculators are **allowed** together with **ONE A4 size double-sided helpsheet.**

# Consultation Dates and timeslots

- **Monday, 25 Oct. : 1400 - 1700 hrs.**  
**Tuesday, 26 Oct. : 1230 - 1430, 1530 - 1700hrs.**  
**Wednesday, 27 Oct. : 0930 - 1230, 1330-1700 hrs.**  
**Thursday, 28 Oct. : 0930 - 1130 hrs.**

**Mass Consult hours on Thurs (non-booking):**  
**1230-1330, 1500-1630 , 1730-1830hrs.**

Additional slots **may** be added, subjected to demand.

- Slots: (**Green - Available, Red - Booked, Cyan - Mass consult**)  
**Consultation Schedule**
- Send me a telegram PM to confirm (30 mins or 1 hour) slots with me.

Default mode will be over Zoom.

Prepare questions/problems to be discussed on the consultation slot.

## probability density function (pdf)

$f(x)$  is said to be a pdf if  $f(x) \geq 0$  and  $\int_{-\infty}^{\infty} f(x) dx = 1$ .

## CDF, Properties of Probabilities in Continuous r.v.

CDF:  $F(x) = \int_{-\infty}^x f(t) dt$  (or equivalently  $\frac{d}{dx}[F(x)] = F'(x) = f(x)$ .)

- $P\{a \leq X \leq b\} = \int_a^b f(x) dx = F(b) - F(a)$ .
- $P\{X = a\} = 0$
- $P\{a \leq X \leq b\} = P\{a < X \leq b\} = P\{a \leq X < b\} = P\{a < X < b\}$

## Expectation of Continuous r.v. $\mathbb{E}(X)$ and $\mathbb{E}(g(X))$

Given pdf  $f(x)$  and real-valued function  $g(x)$ , the expectation  $\mathbb{E}(X)$  and  $\mathbb{E}(g(X))$  are defined as

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f(x) dx \quad \mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x) f(x) dx.$$

if RHS exists.

Define  $P(Z \leq a) = \Phi(a)$ , where  $Z$  follows a Standard Normal distribution.  
Symmetrical properties of Normal Distribution:  $X \sim N(\mu, \sigma^2)$

- $P(X \leq \mu - a) = P(X \geq \mu + a) = 1 - P(X \leq \mu + a)$
- $X$  can be standardised into a Standard Normal distribution by defining

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

- Useful for reading off  $z$ -values:  
 $P(Z \leq a) = P(Z \geq -a) = 1 - P(Z \leq -a)$  or  
 $\Phi(a) = 1 - \Phi(-a)$ .

Note: Standard Normal tables will be provided in the test if necessary.

## De-Moivre Laplace Limit Theorem

Let  $p$  be a constant and  $S_n \sim \text{Bin}(n, p)$ , then for any  $a < b$ ,

$$\lim_{n \rightarrow \infty} \left\{ a \leq \frac{S_n - np}{\sqrt{np(1-p)}} \leq b \right\} = \Phi(b) - \Phi(a)$$

Normal Approximation to Binomial Distribution:  $np(1-p) \geq 10$ .

To approximate  $X \sim \text{Bin}(n, p)$

- $P(X = i) \stackrel{\text{approximation}}{=} P\left(i - \frac{1}{2} \leq X \leq i + \frac{1}{2}\right).$
- $P(X \leq i) \stackrel{\text{approximation}}{=} P\left(X \leq i + \frac{1}{2}\right).$
- $P(X \geq i) \stackrel{\text{approximation}}{=} P\left(X \geq i - \frac{1}{2}\right).$
- $P(X < i) \stackrel{\text{approximation}}{=} P\left(X \leq i - \frac{1}{2}\right).$
- $P(X > i) \stackrel{\text{approximation}}{=} P\left(X \geq i + \frac{1}{2}\right).$

| Distribution               | density / pdf                                                                      | cdf $P(X \leq x)$                                                                  | $\mathbb{E}(X)$          | $\text{Var}(X)$            |
|----------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------|----------------------------|
| Uniform( $a, b$ )          | $\frac{1}{b-a}$<br>$a < x < b$                                                     | $\frac{x-a}{b-a}$<br>$a < x < b$                                                   | $\frac{a+b}{2}$          | $\frac{(b-a)^2}{12}$       |
| Exponential( $\lambda$ )   | $\lambda e^{-\lambda x}, x \geq 0$                                                 | $1 - e^{-\lambda x}$                                                               | $\frac{1}{\lambda}$      | $\frac{1}{\lambda^2}$      |
| Normal( $\mu, \sigma^2$ )  | $\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$                     | $\frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^x e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$ | $\mu$                    | $\sigma^2$                 |
| $Z \sim (0, 1)$            | $\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$                                         | Tables                                                                             | $\mu$                    | $\sigma^2$                 |
| Gamma( $\alpha, \lambda$ ) | $\frac{\lambda^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\lambda t},$<br>$t \geq 0$ | $\frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^x t^{\alpha-1} e^{-\lambda t} dt$    | $\frac{\alpha}{\lambda}$ | $\frac{\alpha}{\lambda^2}$ |

Suppose  $X$  is a random variable with density function  $f_X(x)$  and  $Y$  is a random variable that can be expressed in terms of  $X$ , say,  $Y = g(X)$  for some function  $g$ .

How to determine density function of  $Y$  (or  $f_Y(y)$ )?

- 1 Find the CDF of  $Y$  (or  $F_Y(y)$ ), i.e.,  $P\{Y \leq y\}$ .  
(Express in terms of probability of event about  $X$ )  
In other words, obtain  $P(X \leq h(y))$ .

- 2 Then, differentiate the CDF with respect to  $y$ .

Note: Do look out for the range of values of  $y \in Y$  that satisfies the range of values of  $x \in X$ .



## Joint pmf, marginal pmf

joint probability mass function (pmf)  $p(x, y)$  is

$$p_{XY}(x_i, y_i) = P\{X = x_i, Y = y_i\}.$$

From here, the respective **marginal pmf** of  $X$  and  $Y$  are:

$$p_X(x_i) = \sum_j \underbrace{p_{XY}(x_i, y_j)}_{\text{Sum over all the } y_j} \quad \text{and} \quad p_Y(y_j) = \sum_i \underbrace{p_{XY}(x_i, y_j)}_{\text{Sum over all the } x_i}.$$

## Sum of Independent Random Variables

Suppose that  $X, Y$  are **independent discrete** random variables. Then  $Z = X + Y$  is a discrete random variable with

$$p_Z(z) = \sum_x p_X(x) p_Y(z - x).$$

## (Discrete) Conditional Probability Of $X = x_i$ given $Y = Y_j$

$$p_{X|Y}(x_i|y_j) \\ = P\{X = x_i|Y = y_j\} = \frac{P\{X = x_i \text{ and } Y = y_j\}}{P\{Y = y_j\}} = \frac{p_{XY}(x_i, y_j)}{p_Y(y_j)}.$$

assuming that  $p_Y(y_j) > 0$ .

$p_{X|Y}(x_i|y_j) = p_X(x_i)$  for all  $i, j \iff X$  and  $Y$  are independent.

## Joint continuous CDF and pdf

Joint pdf of  $X, Y$ :  $f(x, y)$ .  $f(x, y) \geq 0$ ,  $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dy dx = 1$ .

Joint CDF:  $F(x, y) = P\{X \leq x, Y \leq y\} = \int_{-\infty}^x \int_{-\infty}^y f(u, v) dv du$ .

Note:  $f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y) = \frac{\partial^2}{\partial y \partial x} F(x, y)$ .

## Marginal pdf

The respective **marginal pdf** of  $X$  and  $Y$  are:

$$f_X(x) = \underbrace{\int_{-\infty}^{\infty} f(x, y) dy}_{\text{Integrate over all the } y} \quad \text{and} \quad f_Y(y) = \underbrace{\int_{-\infty}^{\infty} f(x, y) dx}_{\text{Integrate over all the } x} .$$

Note: State the domain of the marginal pdf.

## Independence of random variables

$X$  and  $Y$  are independent if and only if their joint probability density/mass function can be expressed as

$$f_{XY}(x, y) = h(x)g(y), -\infty < x, y < \infty.$$

where  $h(x)$  **depends only on**  $x$  and  $g(y)$  **depends only on**  $y$ .

Remark: To show that  $X$  and  $Y$  are not independent, it suffices to show that

$$f_{XY}(x, y) \neq f_X(x) \cdot f_Y(y).$$

## Sum of continuous random variables (Convolution)

- For continuous random variables  $X$  and  $Y$ ,  $Z = X + Y$  has the pdf:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z-x) dx.$$

- If  $X$  and  $Y$  are **independent**,  $Z = X + Y$  has the pdf:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x) dx.$$

Evaluating Double integrals:

- Sketch out the 2D region. (Draw out the lines when you see  $x \leq y$  etc.)
- Use corner points to 'test' out the correct region.
- When evaluating things like  $P\{X + Y \leq 1\}$ , convert to  $P\{Y \leq 1 - X\}$  or  $P\{X \leq 1 - Y\}$ . Then, identify the bounds that you need for your integration.

## Sum of Independent Normal variables

If  $X \sim N(\mu_1, \sigma_1^2)$  and  $Y \sim N(\mu_2, \sigma_2^2)$  and  $X$  and  $Y$  are independent, then

$$X + Y \sim (\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

In general, if  $X_i \sim N(\mu_i, \sigma_i^2)$  are mutually independent, then

$$\sum_i X_i \sim N\left(\sum_i \mu_i, \sum_i \sigma_i^2\right).$$

## Conditional distribution

If  $X$  and  $Y$  are jointly continuous random variables, the conditional density of  $Y$  given  $X$  is defined to be

$$f_{Y|X}(y|x) = \begin{cases} \frac{f_{XY}(x,y)}{f_X(x)}, & \text{if } f_X(x) > 0 \\ 0, & \text{otherwise.} \end{cases}$$

Similar for  $f_{X|Y}(x|y)$ , with denominator  $f_Y(y)$ .

Suppose  $X_1, X_2, \dots, X_n$  are **independent** random variables with a common cdf  $F$  and density  $f$ .

Let  $U$  be the **maximum** of the  $X_i$ 's and  $V$  be the **minimum** of the  $X_i$ 's.

- $P(U \leq u) = (F(u))^n$  and  $f_U(u) = n(F(u))^{n-1}f(u)$ .
- $P(V \leq v) = 1 - (1 - F(v))^n$  and  $f_V(v) = n(1 - F(v))^{n-1}f(v)$ .

## Joint Distribution of Max and Min

$$f(u, v) = \begin{cases} 0, & \text{if } u \leq v; \\ n(n-1)f(u)f(v)(F(u) - F(v))^{n-2}, & \text{if } u \geq v. \end{cases}$$

If  $X_i$ 's are uniform on  $[0, 1]$ , then

$$f(u, v) = n(n-1)(u-v)^{n-2} \quad 1 \geq u \geq v \geq 0.$$

# Links to recorded tutorials

Hover over cursor to access the link.

- **Tutorial 1** Password: \*Lu#+9f&
- **Tutorial 4** Password: f9US\*i81
- **Tutorial 5** Password: 8w?97Wj1
- **Reviews 1 and 2** Password: J=!eg+7S
- **Tutorial 6** Password: z#j55rgA
- **Tutorial 7** Password: 9ww6s###\$
- **Tutorial 8** Passcode: yCik&3Rb